

MODEL QUESTION PAPER

TED (15)5093

Reg No.....

(RIVISION – 2015)

Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

CHEMICAL ENGINEERING
(Maximum Marks: 100)

Time:3Hrs

PART - A

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define grammole?
2. Define density and specific gravity?
3. Define thermal conductivity?
4. State Fick's law of Diffusion?
5. Define Relative volatility? (5X2=10)

PART - B

(Maximum Marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Determine the amount of Na_2CO_3 required to prepare 250ml 0.5N solution?
2. Describe the various Systems of Units?
3. Derive the equation to calculate heat transfer through spherical wall?
4. Describe Convection & types of Convection?
5. Explain the factors controlling the rate of absorption?
6. Describe Liquid- Liquid extraction, the terms Raffinate and Extract?
7. Describe Fractional Distillation? (5X6=30)

PART -C

(Maximum Marks: 60)

(Answer one full question from each MODULE. Each full question carries 15 marks)

MODULE -I

- III a 0.8 grams of NaOH is dissolved in 100ml water. Calculate its Molarity and the molar volume at 2atm pressure and 100°C? (8)
- b. Using Dimensional Analysis convert Poise into Centipoise? (7)

OR

- IV a. Write the formula of Oxalic acid and determine its molecular weight and equivalent weight? (8)
- b. Explain the difference between Fundamental and Derived Units? (7)

MODULE –II

- V a. Explain the heat transfer in boiling liquid and regimes of boiling? (8)
- b. Explain conduction and Fourier's law of heat conduction? (7)

OR

- VI a Explain the mechanism of Natural Convection? (7)
- b. A furnace constructed with 100mm of thick fire brick, 50mm insulating brick and 150mm of the building brick. The inside and outside temperatures are 1200°K & 500°K respectively. Find the heat loss per unit area?" k" values are 1.8, 0.4, 0.9W/m ° K for fire brick, insulating brick& building brick respectively (8)

MODULE -III

- VII a. Explain the mechanism of Absorption and Henry's law? (8)
- b. Explain Leaching? (7)

OR

- VIII a. Explain the working of Horizontal Bollman Extractor? (8)
- b. Explain Adsorption? (7)

MODULE - IV

IX a. Explain ideal and Non ideal solutions? (7)

b. Describe Maximum and Minimum Boiling Azeotropes? (8)

OR

X a. Explain steam Distillation? (7)

b. Explain Raoult's law and derive its equation? (8)